Question

Give the products $\mathbf{A}$ of the reaction in Figure 1 and the product $\mathbf{B}$ of the reaction in Figure 2.

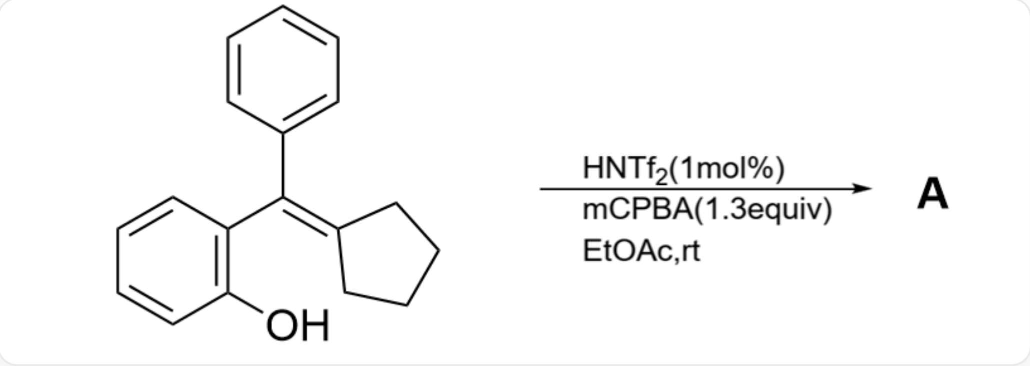

Fig. 1, the reaction in the figure is described by SMILES as: OC(C=CC=C1)=C1/C(C2=CC=CC=C2)=C3CCCC\3>>[[A]], where the reaction conditions are $\mathrm{HNTf_2(1mol\%), mCPBA(1.3equiv), EtOAc, rt}$

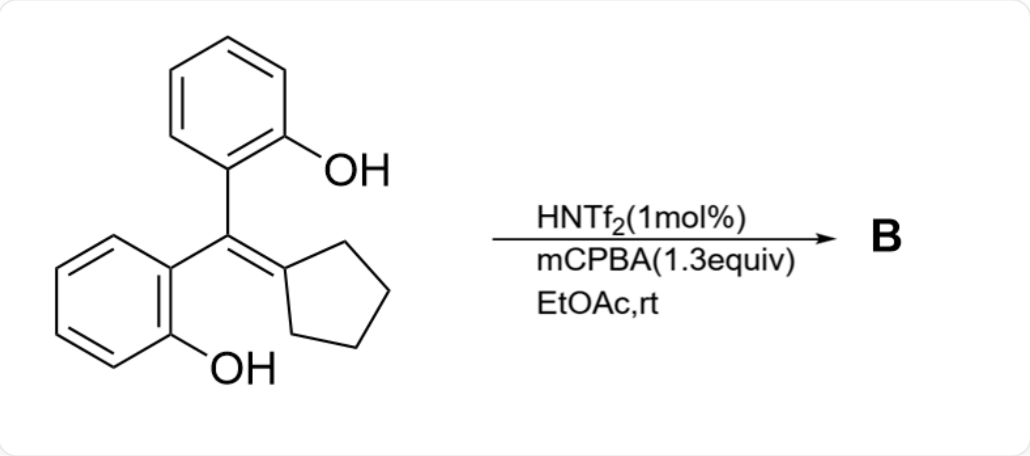

Fig. 2, the reaction in the figure is described by SMILES as: OC(C=CC=C1)=C1/C(C2=C(O)C=CC=C2)=C3CCCC\3>>[[B]], where the reaction conditions are $\mathrm{HNTf_2(1mol\%), mCPBA(1.3equiv), EtOAc, rt}$

The following statements are made:

1. Only one carbon-carbon single bond cleavage occurs during the formation of both $\mathbf{A}$ and $\mathbf{B}$.

2. **A** has three rings of six members or less.

3. **B** has five rings of six members or less.

4. The most important reason why the products of Figure 1 and Figure 2 are different is the electron richness of the benzene ring.

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: G

Detailed Explanation

Figures 1 and 2 share the same principle in the first half of the reaction: mCPBA introduces an epoxy group onto the reactant double bond.

CHECKPOINT 1 PTS

mCPBA introduces an epoxy group onto the reactant double bond

$\mathrm{HNTf_2}$ is a strong acid, protonating the epoxy group and dissociating into a hydroxyl group, generating the most stable carbocation on the carbon with simultaneous conjugation stabilization of the two benzene rings.

CHECKPOINT 1 PTS

$\mathrm{HNTf_2}$ catalyzes the opening of the epoxide, forming a carbocation at the benzylic position

This intermediate structure is similar to the structure after the departure of a hydroxyl group in the Pinacol rearrangement, so a concerted carbon-carbon bond migration occurs in the adjacent position, yielding the intermediate in Figure 3 or the intermediate in Figure 4.

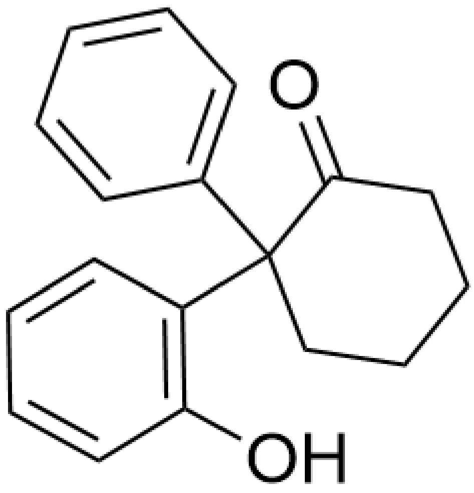

Fig. 3, the molecule in the figure is described by SMILES as: OC1=C(C(CCCC2)(C3=CC=CC=C3)C2=O)C=CC=C1

CHECKPOINT 1 PTS

The carbocation intermediate undergoes a Pinacol-like rearrangement, and the intermediate generated in the reaction of Figure 1 is represented by SMILES as: OC1=C(C(CCCC2)(C3=CC=CC=C3)C2=O)C=CC=C1

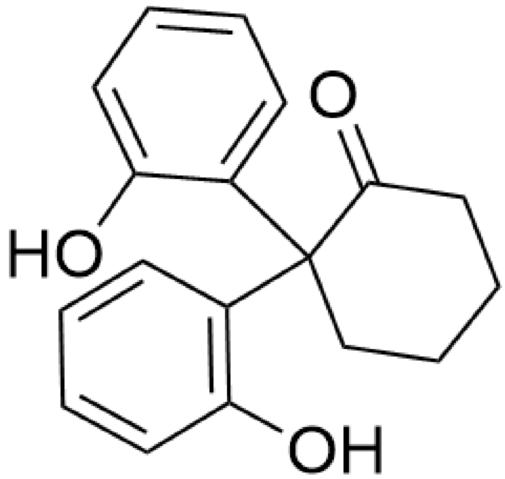

Fig. 4, the molecule in the figure is described by SMILES as: OC1=C(C(CCCC2)(C3=CC=CC=C3O)C2=O)C=CC=C1

CHECKPOINT 1 PTS

The carbocation intermediate undergoes a Pinacol-like rearrangement, and the intermediate generated in the reaction of Figure 2 is represented by SMILES as: OC1=C(C(CCCC2)(C3=CC=CC=C3O)C2=O)C=CC=C1

The carbonyl group of the intermediate can readily form an acetal or hemiacetal structure with the intramolecular hydroxyl group, yielding the final product $\mathbf{A}$ in Figure 5 or the final product $\mathbf{B}$ in Figure 6.

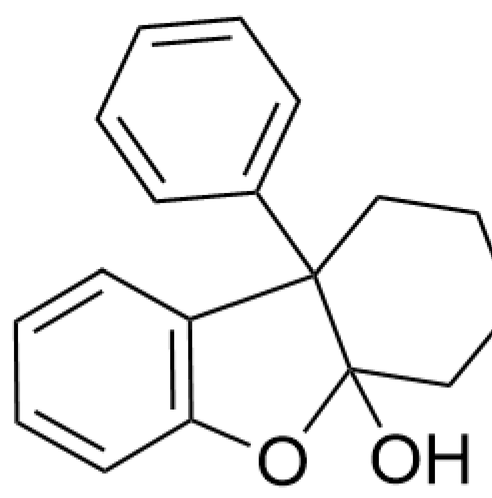

Fig. 5, the molecule in the figure is described by SMILES as: OC12CCCCC1(C3=CC=CC=C3)C4=C(O2)C=CC=C4

CHECKPOINT 1 PTS

The reaction in Figure 1 forms an intramolecular hemiacetal structure, and the final product **A** is represented by SMILES as: OC12CCCCC1(C3=CC=CC=C3)C4=C(O2)C=CC=C4

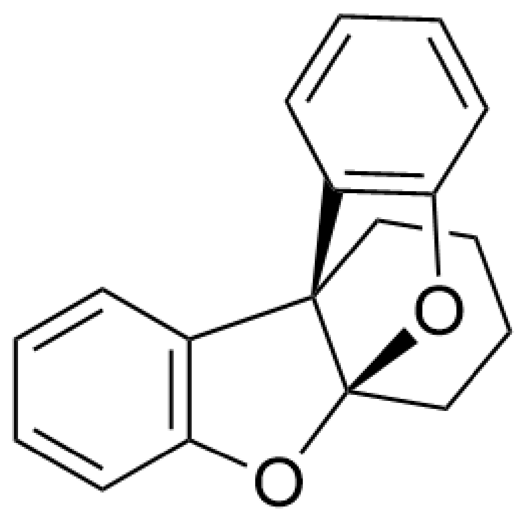

Fig. 6, the molecule in the figure is described by SMILES as:
C1([C@]2(C3=C(O4)C=CC=C3)CCCC[C@]24O5)=C5C=CC=C1

CHECKPOINT 1 PTS

The reaction in Figure 2 forms an intramolecular hemiacetal structure, and the final product $\mathbf{B}$ is represented by SMILES as: C1([C@]2(C3=C(O4)C=CC=C3)CCCC[C@]24O5)=C5C=CC=C1

According to the above mechanism, only one carbon-carbon single bond cleavage occurs in the process of generating $\mathbf{A}$ and $\mathbf{B}$, so statement 1 is correct.

$\mathbf{A}$ has four rings with six or fewer members, so statement 2 is incorrect.

$\mathbf{B}$ has five rings with six or fewer members, so statement 3 is correct.

The most important reason why the products of Figure 1 and Figure 2 are different is that the last step forms an acetal or hemiacetal, which is not related to the electron-richness of the benzene ring.